DS-2002: Data Systems

Overview of Data Warehouse Systems

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Modern Data Platform: Solution Scenarios

Big (Unstructured and/or Poly-Schematic) Data Integration and Advanced Analytics

"We want to integrate all our data into our data warehouse"

Modern Data Warehousing

"We're trying to predict which of our customers will churn"

Advanced Analytics

"We're trying to get insights from our devices in real-time"

Real-Time Analytics

Modern Data Platform: Data Services Pipeline

INGEST

Data Orchestration & Monitoring

Data Integration

STORE

Unstructured Data Storage

Data Lake Storage

PREPARE

Data Cleansing, Transformation & Streaming

Databricks

SERVE

Data Modeling, Serving & Storing

Data Warehouse

PREDICT

Reporting, BI, Predictive Analytics & AI

Machine Learning

DELIVER

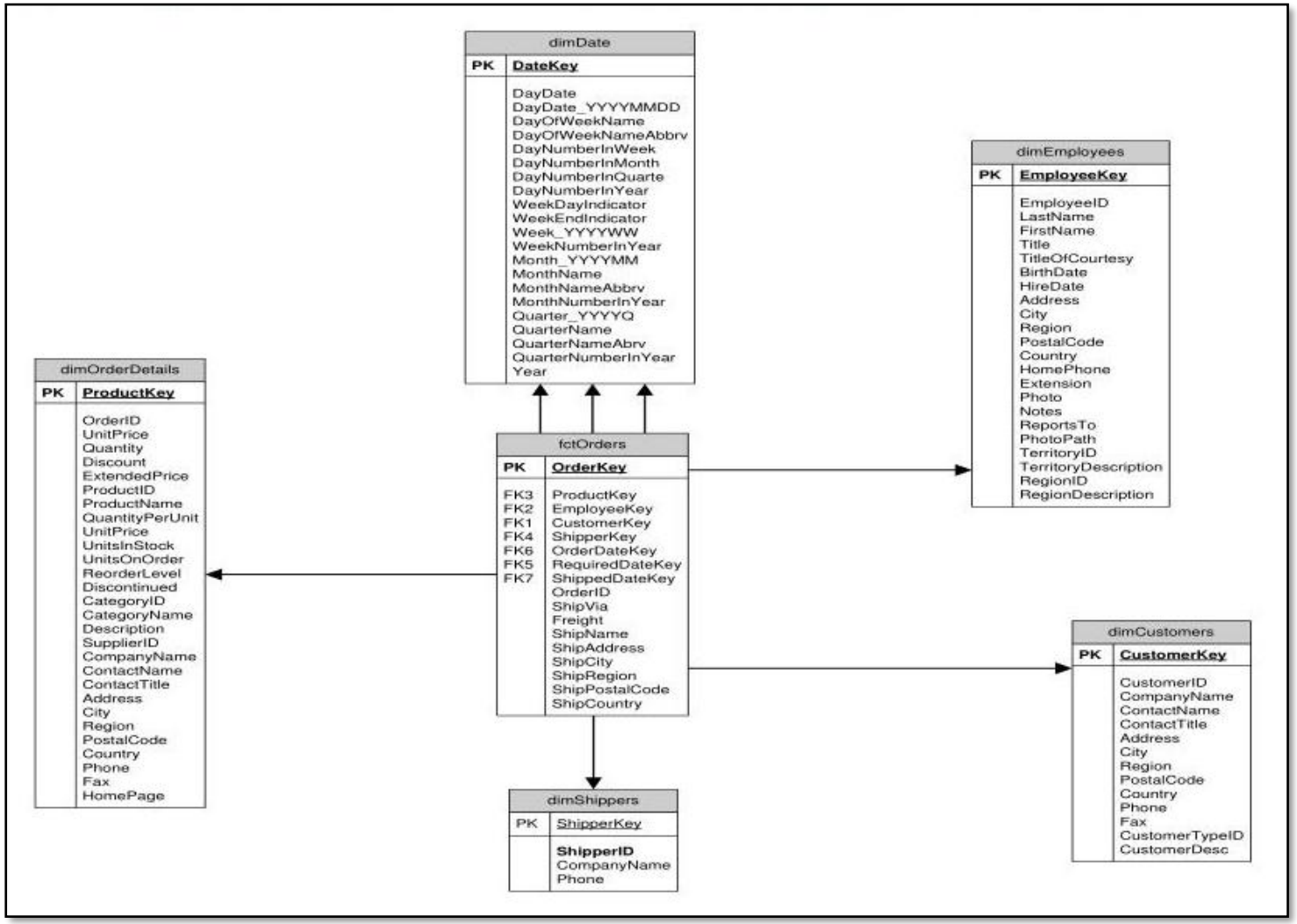
MLOps Integration: CI/CD Pipelines, Version Control, Monitoring, Test Automation, Infrastructure as Code, Containers, Microservices

DevOps

The Data Warehouse Process

How to Approach Designing and Building a Data Warehouse

OLAP Database: Multi-Dimensional (Star) Schema



The Four-Step Dimensional Design Process

A Time-Honored and Tested Methodology for Delivering Data Marts & Data Warehouses

1. Select the Business Process

- Prioritize Business Processes that Have Significant Business Impact, and that can be Easily Delivered
- Attempting to Boil-the-Ocean Increases the Risk of Project Failure

2. Declare the Grain

- Determine What Level of Detail Must Be Available
- Greater Atomicity Provides Greater Analytic Flexibility
- Aggregated Data Limits Access to Transaction Detail

3. Identify the Dimensions

- Categorical Values Describing Each Business Process
- Defines How Transactions are Viewed or Filtered
- Should Have the Same Meaning Across Processes

4. Identify the Facts

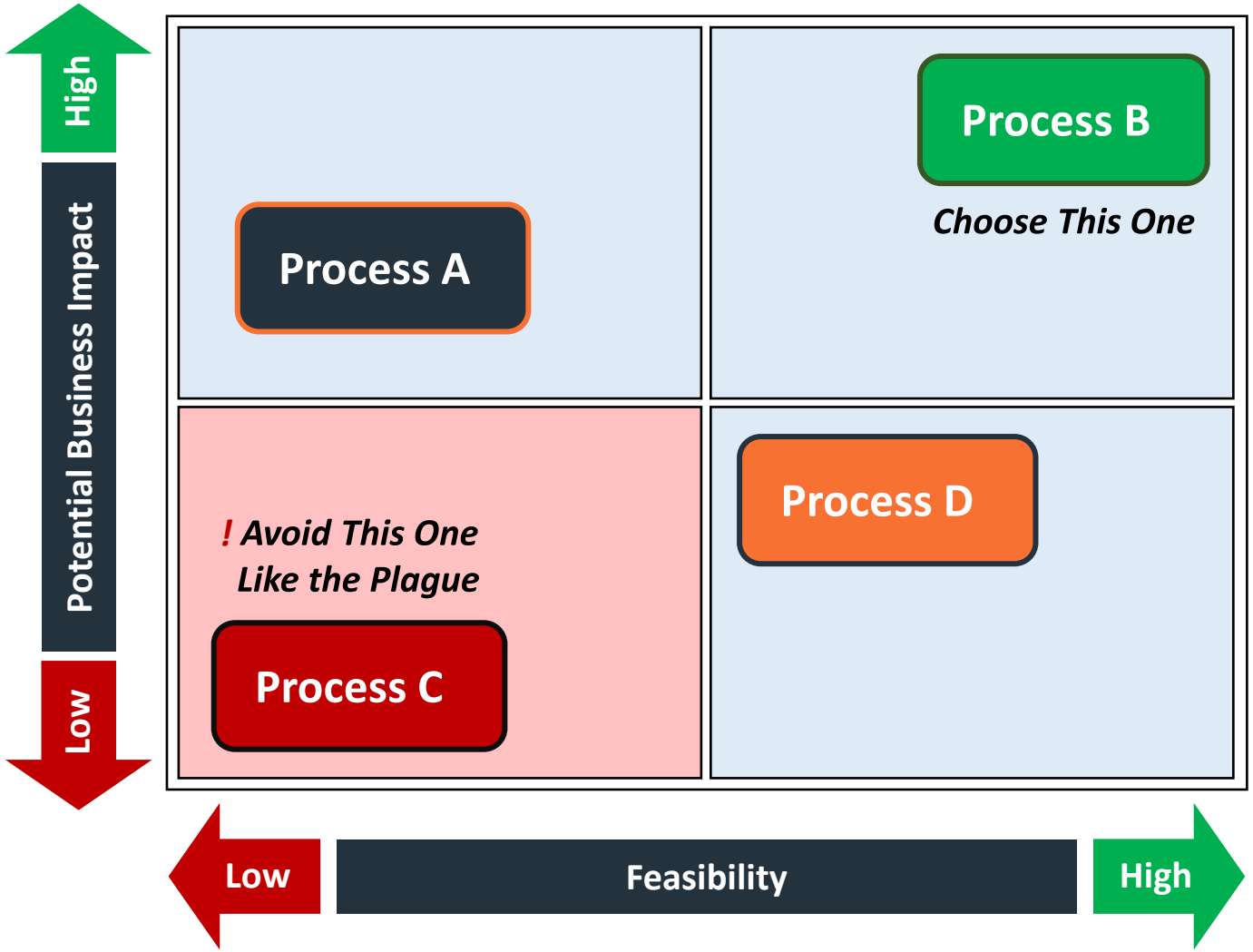
- Numeric Values that Quantify Each Business Process
- Defines How Transactions Are Summarized
- Facts Must Be True-to-the-Grain

The Data Warehouse Toolkit, by Ralph Kimball

Selecting Business Processes: Prioritizing Requirements

Quadrant Analysis for Prioritizing Requirements:

- **Business Process A:**
  - High Potential Business Impact
  - Extremely Difficult to Implement
- **Business Process B:**
  - **High Potential Business Impact**
  - **Highly Feasible**
- **Business Process C:**
  - Very Little Business Impact
  - Extremely Difficult to Implement
- **Business Process D:**
  - Little Business Impact
  - Highly Feasible



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Identifying Dimensions: Data Warehouse Bus Matrix

Using the Same Dimensions Across Multiple Business Processes Enforces a Unified View of the Truth

	Common *(Conformed)* Dimensions			
Business Processes	**Date**	**Product**	**Store**	**Promotion**
Sales	✓	✓	✓	✓
Inventory	✓	✓	✓	
Deliveries	✓	✓	✓	

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Business Processes	Common *(Conformed)* Dimensions			
	Date	Product	Store	Promotion
Sales	✓	✓	✓	✓
Inventory	✓	✓	✓	
Deliveries	✓	✓	✓	
WH Inventory	✓	✓		
WH Deliveries	✓	✓		
Purchase Orders	✓	✓		

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Identifying Dimensions: Data Warehouse Bus Matrix

Using the Same Dimensions Across Multiple Business Processes Enforces a Unified View of the Truth

Business Processes	Common *(Conformed)* Dimensions							
	Date	Product	Store	Promotion	Warehouse	Vendor	Contract	Shipper
Sales	✓	✓	✓	✓				
Inventory	✓	✓	✓					
Deliveries	✓	✓	✓					
WH Inventory	✓	✓			✓	✓		
WH Deliveries	✓	✓			✓	✓		
Purchase Orders	✓	✓			✓	✓	✓	✓

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Database Transactions: Enforcing Data Integrity

The **ACID** Properties of a Transaction Must be Enforced to Ensure the Integrity of the Data

Atomicity

- All operations in a transaction must be treated as a single [atomic] unit, and must either be fully committed (i.e., succeed) or be rolled-back (i.e., have no effect whatsoever).

Consistency

- Each transaction must guarantee the valid state of the database; both before and after the transaction is either committed or rolled-back.
- All defined rules, constraints, and relationships must be enforced and maintained.

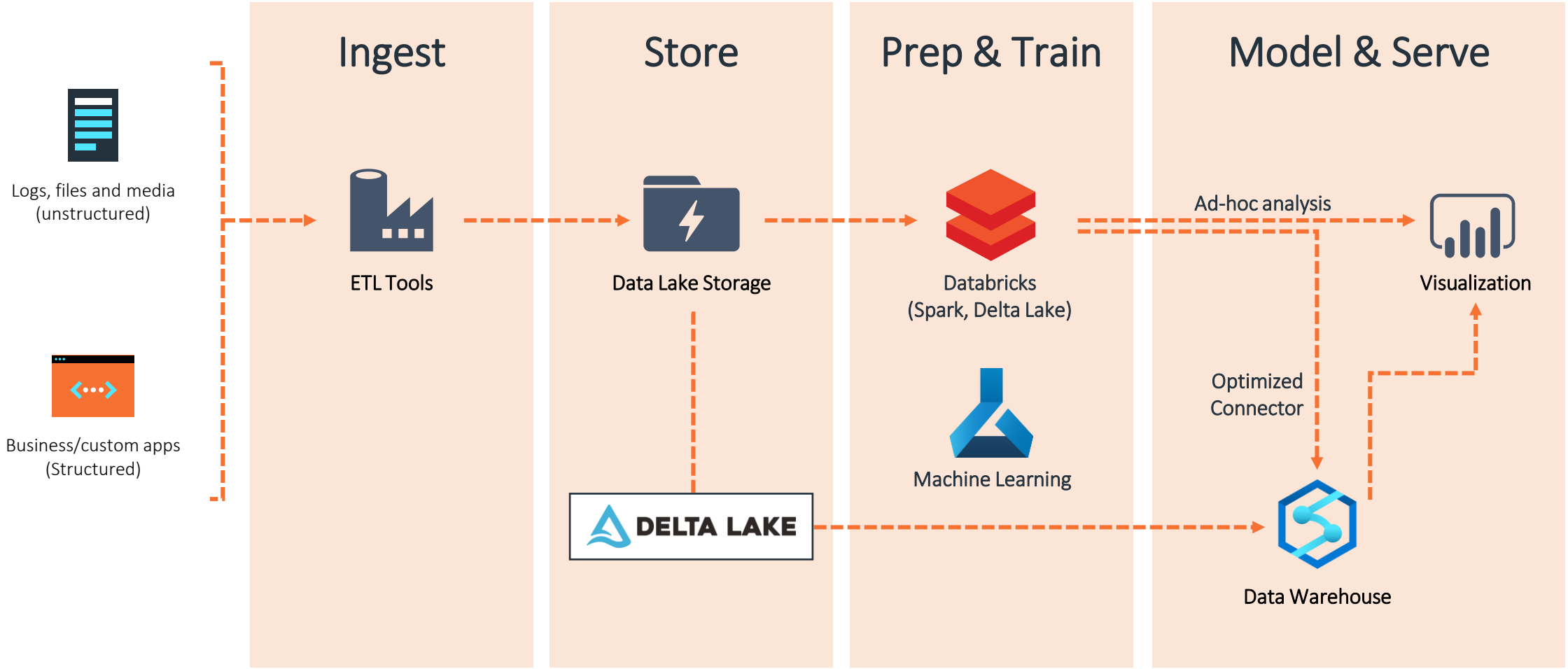
Isolation

- Concurrent transactions must be completely independent; having no effect on each other.
- Transactions must remain invisible to each other until they are either committed or rolled-back.

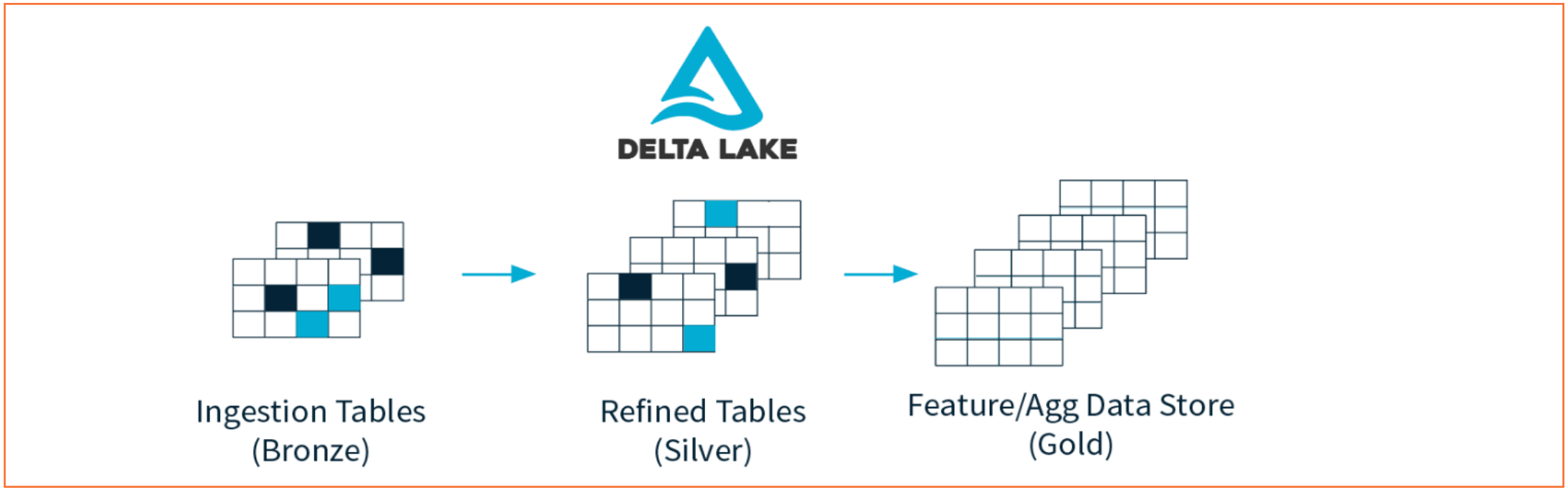
Durability

- Once committed, the effects of each transaction must be permanently saved (written to disk); even if the system fails amidst their execution.
- A transaction log (aka, Journal file) serves as a write-ahead mechanism to ensure each committed transaction can be replayed to successfully recover the database to a consistent state in the event of a system failure.

Data Engineering… for Data Science



Databricks: Delta Lake at Scale



ACID Transaction Guarantees

Atomic, Consistent, Isolated, Durable

Versioned Parquet Files

Delta transaction log keeps track of all operations

Efficient Upserts

MERGE, DELETE, UPDATE

Time Travel

Audit history, pipeline debugging, data reproducibility

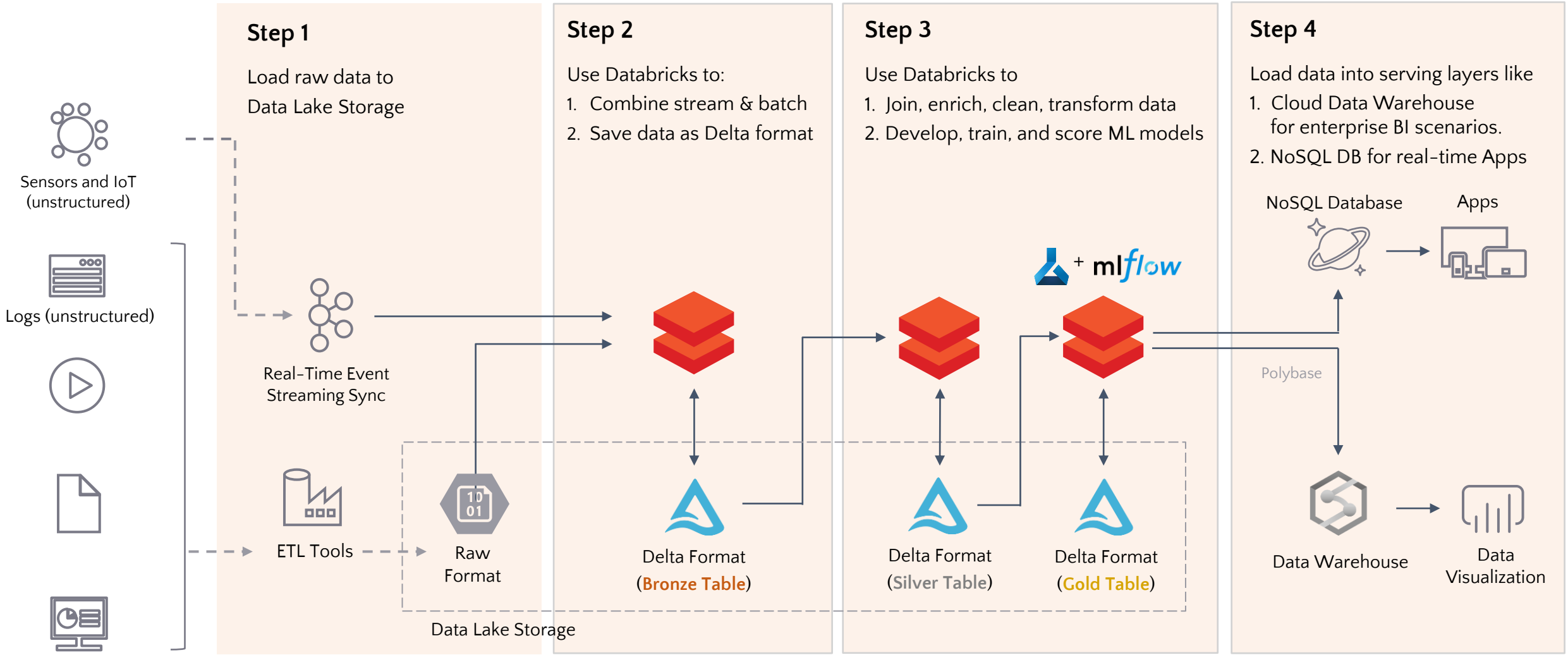
Small file compaction with no interrupt to availability

OPTIMIZE and VACUUM

Z-Order partitioning with up to 100x perf

New multidimensional partitioning enables data skipping

Design Pattern: Modern Data Warehousing



Q & A

An Overview of Data Warehouse Systems